KONERU LAKSHMAIAH EDUCATION FOUNDATION
Database Systems Engineering and Distributed Backend Development (25CS1302E)

Trimester 4 · 2026–2027
Section 7 · Team 3

Smart Logistics & Delivery Management · Indexed SQL + Distributed Architecture

LOGIFLOW

Smart Delivery Management System

Track · Manage · Deliver

A web-based platform that simplifies and automates shipment management, tracking and route planning.

COURSE	COURSE CODE	SECTION · TEAM	BUILT WITH
Database Systems Engineering and Distributed Backend Development	**25CS1302E** Trimester 4 · 2026–2027	**Section 7** Team 3	**Flask · SQL · DBMS** HTML5 · CSS3 · JavaScript

THE PROBLEM

Manual delivery handling causes **late shipments**, **scattered records** and **slow lookups** as operations grow across locations.

- ! **Late shipments**
- ! **Scattered records**
- ! **Slow lookups at scale**

OBJECTIVE

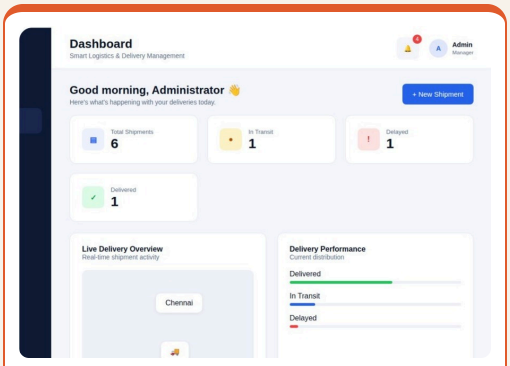
Reduce delivery delays, improve transportation efficiency and give logistics administrators clear visibility of every shipment.

- ✓ **Fewer delivery delays**
- ✓ **Efficient transportation**
- ✓ **Full shipment visibility**

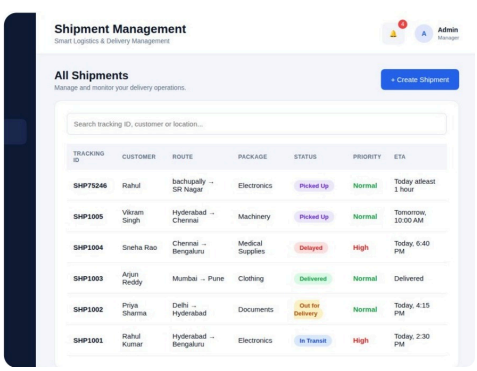
KEY FEATURES

1. **Live Dashboard**
2. **Shipment Management**
3. **Live Tracking & ETA**
4. **Smart Notifications**
5. **Query Analytics**

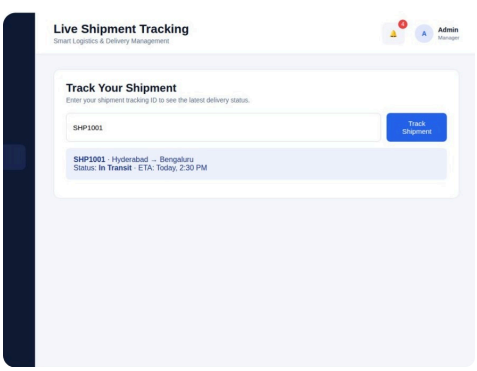
SYSTEM SCREENSHOTS



Dashboard
Live counts of total, in-transit, delayed and delivered shipments.



Shipment Management
Create, search and monitor every delivery in one table.



Live Tracking
Enter a tracking ID to see status and ETA instantly.

DATABASE LAYER · SQL & INDEXING

SELECT, INSERT, UPDATE, DELETE, JOIN and GROUP BY manage shipments, customers and delivery records.

`SELECT` `INSERT` `UPDATE` `DELETE` `JOIN` `GROUP BY`

QUERY PERFORMANCE · FAST LOOKUPS

Tracking ID Search	8 ms
Customer Search	12 ms
Delivery History	21 ms

DISTRIBUTED ARCHITECTURE

Replication · Read scaling · Sharding · CAP theorem

Primary Database Hyderabad → **Replica 01** · Bengaluru, **Replica 02** · Mumbai

BUILT WITH

HTML5 · CSS3 · JavaScript · Python (Flask)
SQL · DBMS · VS Code

TEAM

Konda Vaishali · 2520030117
P. Bindhu Madhavi · 2520030077
Alaparthi Nava Harsha · 2520030493